Supervisor Meeting Brief: Where the Thesis Stands and What to Decide

Regime-Switching Risk Models for ETF Portfolios Prepared for Prof. Seo. All results use data through 2019-12-31 only; 2020-2025 has not been looked at.

1. Bottom line (read this first)

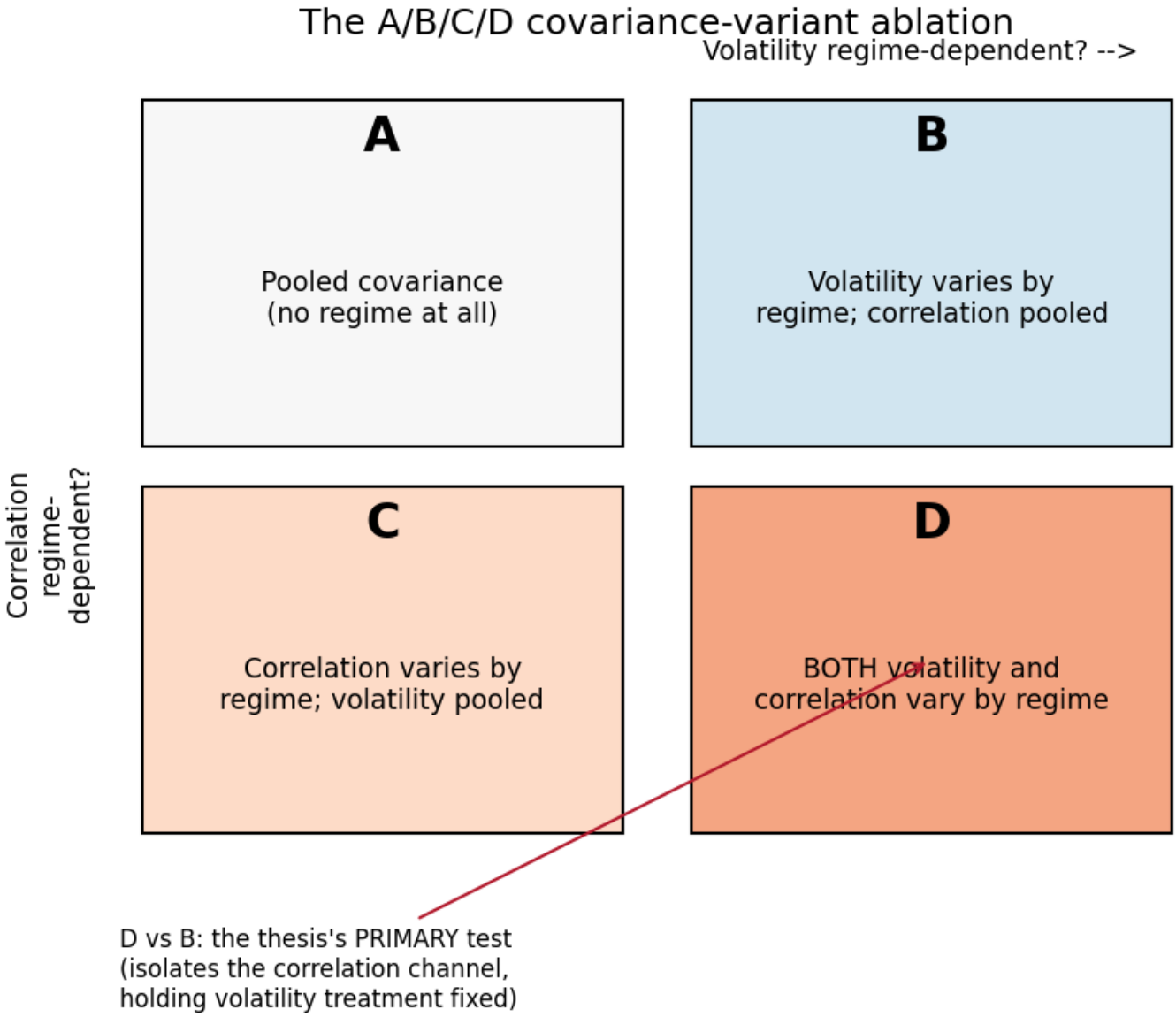
1. **The pilot said "proceed", and it was right to check first.** All six pre-registered go/no-go criteria passed. The pilot also showed the test is weak: with about 10 years of daily data it detects only a *strong* correlation shift (about 50% power), and a moderate one only about 25% of the time.
2. **Broad exploration (as you asked) found no evidence that regime-switching *correlations* improve risk forecasts.** Across 10 ETF universes, 10 ways to define a regime, and a 28-year stock/bond sample, letting correlations switch gave no detectable gain over letting only volatilities switch (0 of 214 tests significant after correction).
3. **Regime models also lose to simple smooth models.** No regime model we tried beat GARCH-t or DCC-GARCH on tail-risk forecasts (0 of 214 comparisons).
4. **The thesis's motivating case is missing from the data we are allowed to use.** Stocks and bonds moved *together* (positive correlation) on 91% of days in 1988-1999 but on 0% of days in 2010-2019. The 2022 reversal sits in the held-out 2020-2025 data.
5. **Recommendation:** reframe the thesis as an honest *detectability* study with **one pre-registered confirmatory test on 2020-2025**. Section 6 lists the four decisions needed from you.

2. The question and the theory

Question. For a fixed diversified ETF portfolio, does a real-time regime-aware model give better one-day-ahead tail-risk forecasts (Value-at-Risk, VaR, and Expected Shortfall, ES) than conventional models?

Regimes. A 2-state hidden Markov model (HMM) reads daily returns and outputs, each day, the probability that the market is in "calm" or "stress". Every forecast for day t+1 uses only information up to day t (no look-ahead).

The ablation (the core design). Portfolio risk depends on volatilities and correlations. We let each switch by regime separately:



- **B vs A** asks whether regime-dependent *volatility* helps.
- **D vs B is the thesis's primary test.** It holds the volatility treatment fixed and asks whether letting *correlations* switch adds anything. This is the part of the design that is new.
- **Benchmarks:** historical simulation, EWMA, GARCH-t, and DCC-GARCH (a smooth, non-regime model of time-varying correlation).

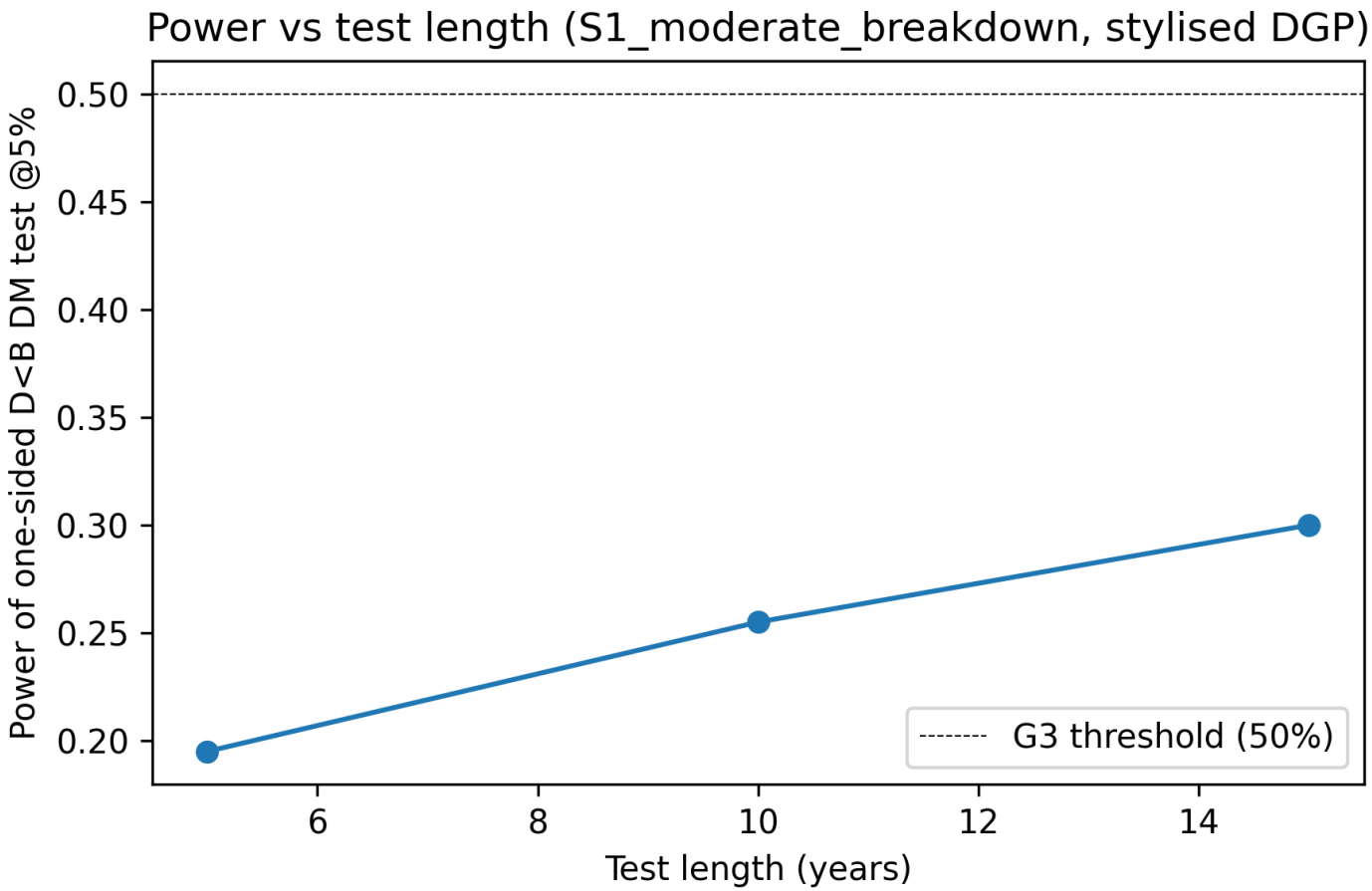
Scoring. The FZ0 loss scores VaR and ES jointly (lower is better). Diebold-Mariano (DM) tests compare two models' losses. A Holm correction handles the many comparisons.

Why the correlation channel matters in theory. Diversification fails if correlations jump in stress. But correlations also rise mechanically when volatility rises (Forbes and Rigobon, 2002), so a measured jump is partly an artefact, which is one reason this channel is hard to isolate.

3. Pilot results (SPY / IEF / GLD, equal weight)

Check	Result
Code correctness	41 of 41 tests pass
False-positive rate of the D-vs-B test	5.0% (stylised) / 2.0% (calibrated), within the 10% limit
Power, strong correlation shift, 10 years	52.5% (stylised) / 42.0% (calibrated)

Power, moderate shift, 10 years	25.5% / 16.5%
Real data 2010-2015, D vs B	t = -0.47, p = 0.64: not significant
Regime stability	90.3% of monthly refits are well behaved (pass, but marginal)
Reproducibility	Identical on rerun



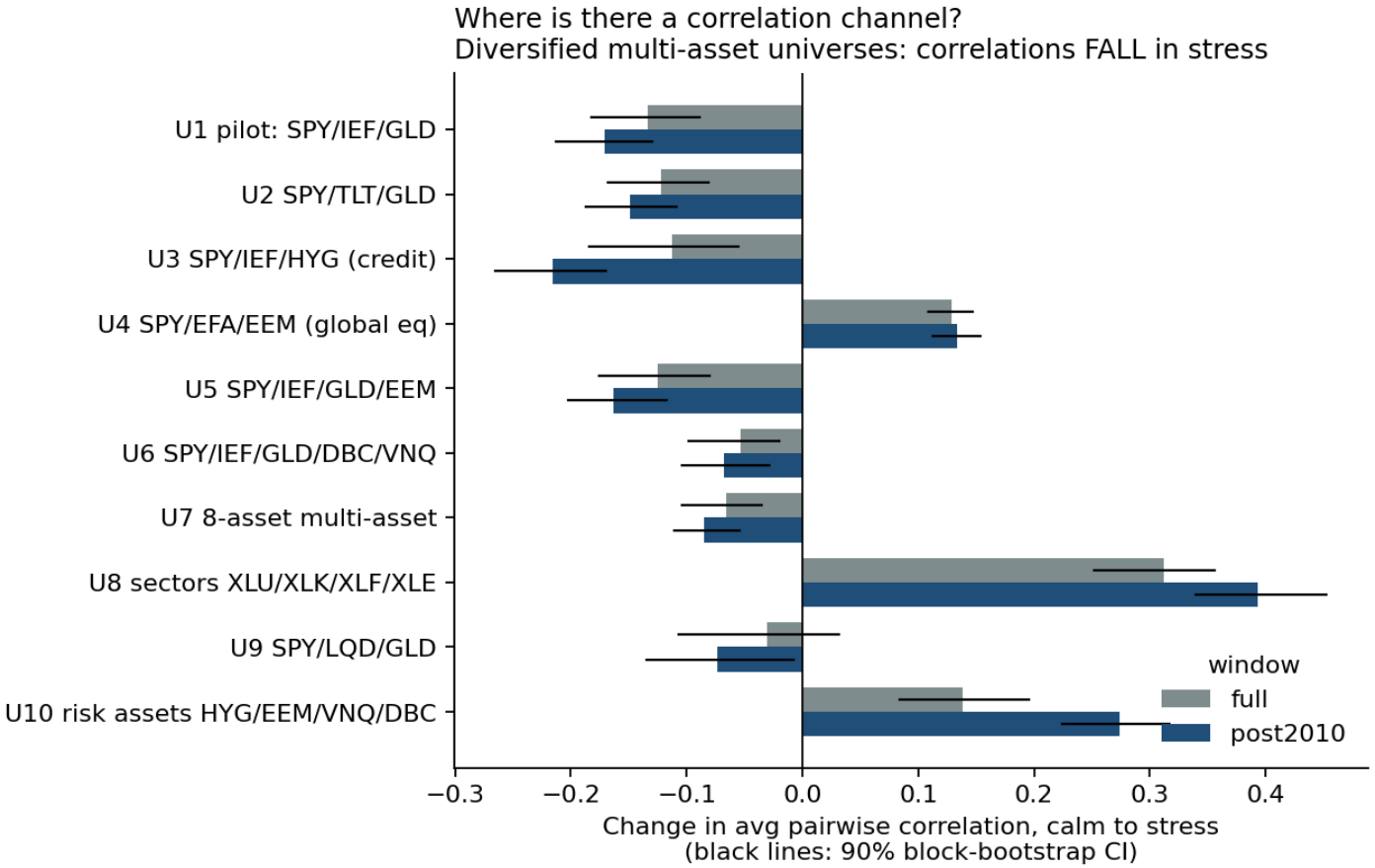
Reading. More data helps slowly: the moderate-shift power only reaches 30% at 15 years. A null result on this design is therefore weak evidence of "no effect".

4. Exploration results (what you asked us to try)

Exploration covered 10 universes, 7 descriptive regime definitions, 10 backtested regime signals (including bond-aware ones), and a 1986-2019 fund proxy sample. Everything tried is logged in `output/explore/`. The results are descriptive and in-sample.

Finding 1: In a diversified portfolio, correlations fall in stress

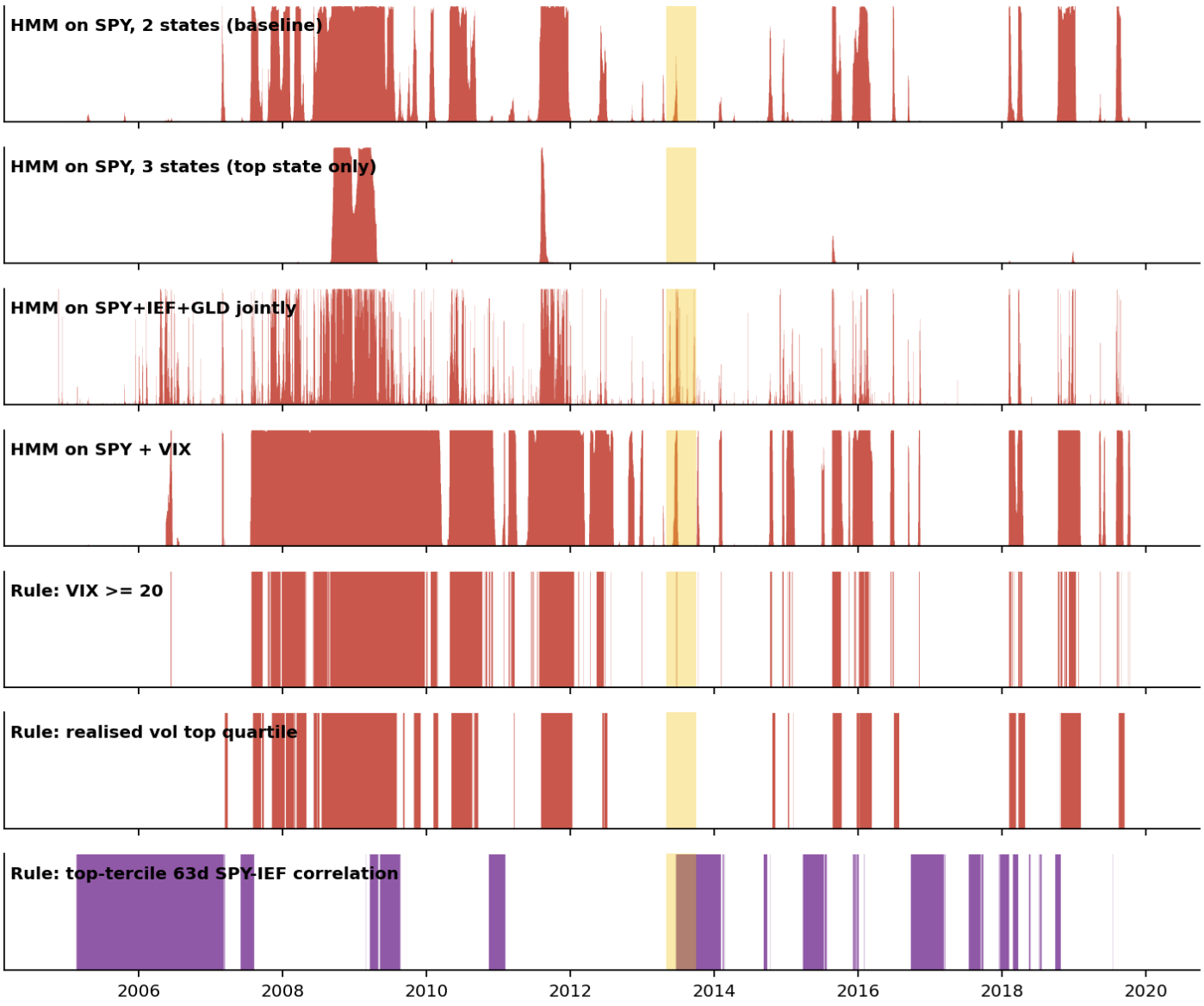
Treasuries become *more* negatively correlated with stocks in stress, which cancels the rise among risk assets. Only equity-only or risk-asset universes (global equity, sectors, HYG/EEM/VNQ/DBC) show a correlation increase.



Finding 2: The definition of "stress" barely matters, and all volatility-based definitions miss the 2013 taper tantrum

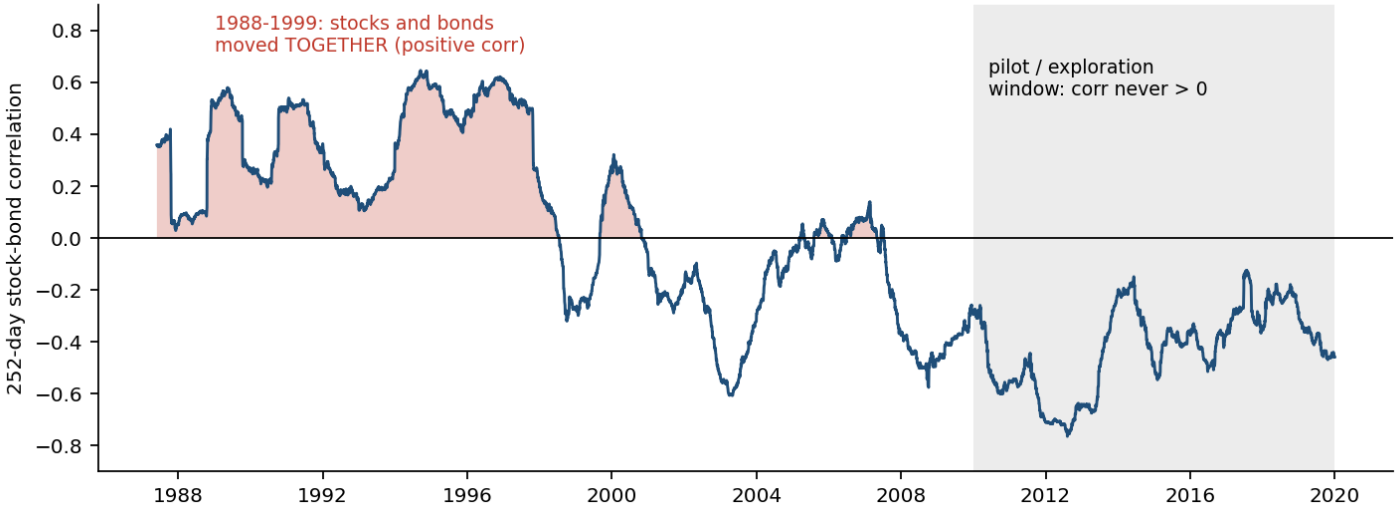
Four volatility-based definitions give the same picture. The 2013 taper tantrum was a bond-led shock that equity volatility did not flag. Only a correlation-defined regime flags it (yellow band below), but it is a *different* regime: it also fires in calm 2017 and ignores the 2008 crisis.

Seven ways to define 'stress' on SPY/IEF/GLD (yellow band = 2013 taper tantrum)
All volatility-based definitions agree and all miss the taper tantrum; only the correlation-defined regime flags it



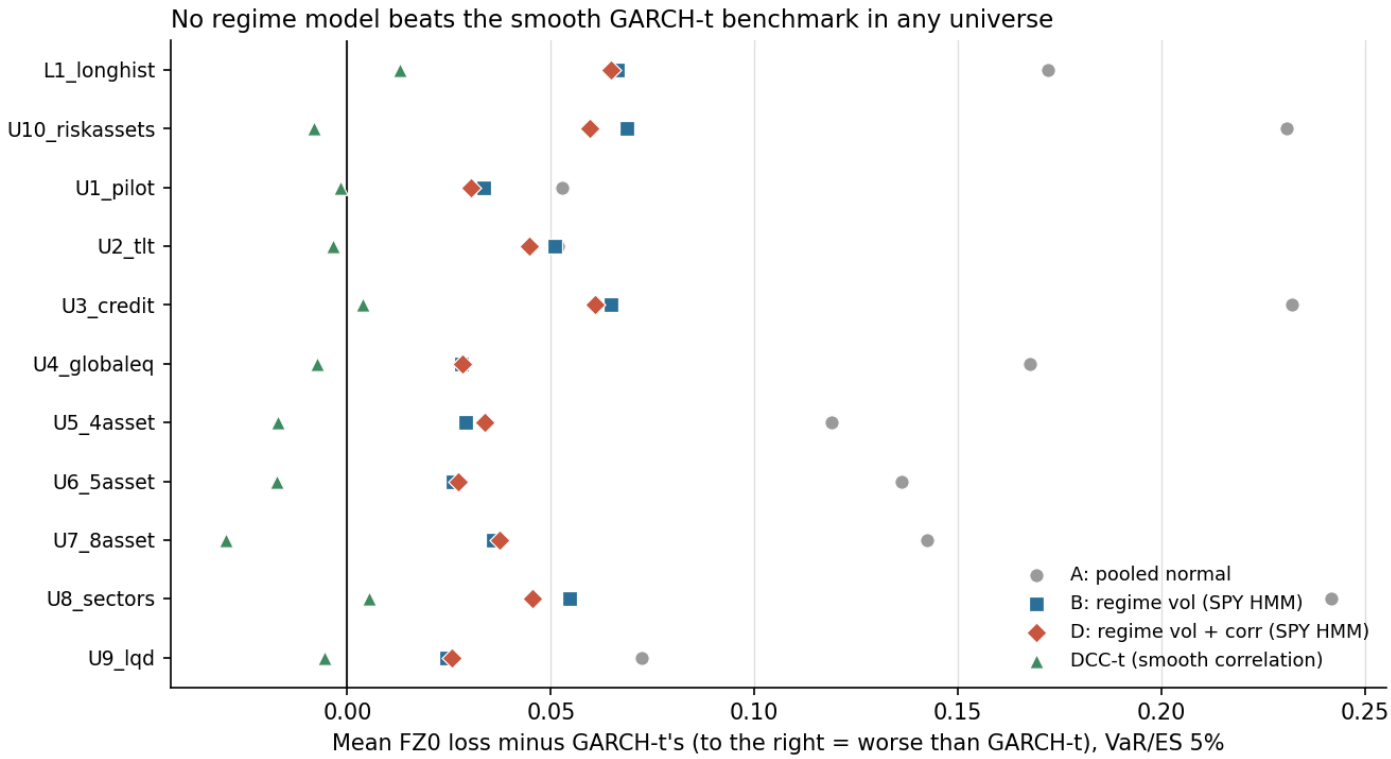
Finding 3: The positive stock-bond correlation that motivates the thesis is absent from 2010-2019

A positive stock-bond correlation regime exists in history, but not in 2010-2019
(VFINX vs VUSTX, S&P 500 vs long Treasury fund; 2020+ held out)



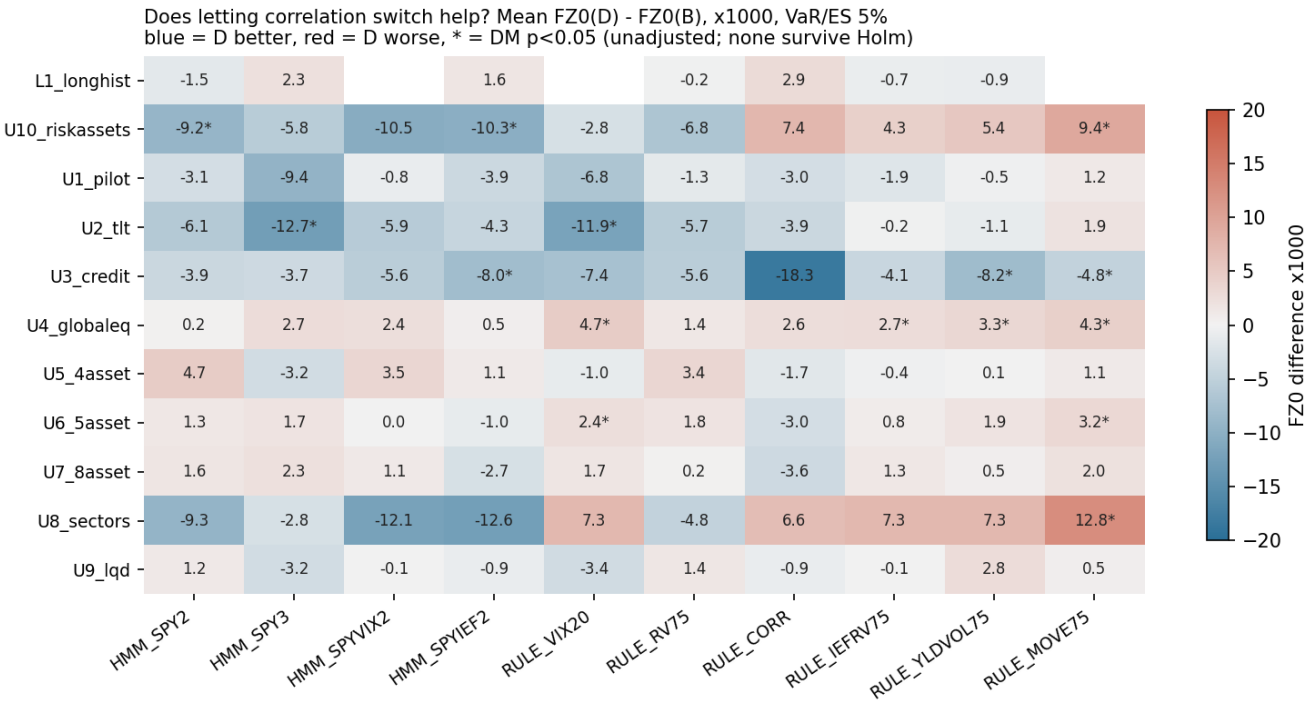
Share of days with positive correlation: **91%** in 1988-1999, **24%** in 2000-2009, **0%** in 2010-2019. Anything we develop on modern ETFs never sees this regime.

Finding 4: Regime models beat a pooled normal but never beat smooth benchmarks



- Regime-switching covariance beats the pooled normal (model A) significantly in 105 of 214 comparisons.
- It is never significantly better than GARCH-t or DCC-t (0 of 214) and is significantly worse in 96 and 124 of the 214, respectively.
- GARCH-t and DCC-t are statistically indistinguishable in 9 of 11 samples. The smooth-correlation benchmark does the "correlation" job as well as, or better than, a regime switch.

Finding 5: The correlation channel (D vs B) is not detectable



- 37 of 214 tests have raw $p < 0.05$, against about 11 expected by chance. They split 18 favouring D and 19 favouring B. **None survive the Holm correction.**
- The sign follows the signal: equity-volatility HMMs favour D in equity-heavy universes, while bond-aware rules (MOVE, yield volatility, IEF volatility) favour B there.
- **Even in 1992-1999**, when stocks and bonds moved together, D-B is within noise for six of seven signals.
- The sample is underpowered: the smallest detectable FZ0 difference is about 0.009, while the typical observed difference is about 0.003.
- Bond-aware signals (direction B below) gave no consistent gain over equity-volatility signals.

5. What this means

- **We cannot claim that regime-switching correlations help.** With these data and this design we also cannot claim they do not, because the test is underpowered.
- **The smooth DCC-GARCH benchmark is the model to beat**, and no regime model beats it. A thesis that omits it would overstate the case for regimes.
- **The one episode that could discriminate is 2022.** It is a single, slow structural shift rather than a fast regime, so even the confirmatory test will have limited power. This has to be said up front.

6. Directions and decisions needed from you

	Direction	What we found	Suggested priority
D	Report power and detectability as the headline result, with the null stated plainly	Matches every result above; a safe fallback	**Primary**
Confirmatory test	One fixed specification, run once on 2020-2025	Needs your approval before anything touches that data	**Do next**
A	Extend the development sample back to 1986 using fund proxies	Done as a check: does not rescue the correlation channel	Supplementary only
B	Add bond-aware or correlation-defined regime signals	Done: no consistent gain; mechanical overlap between "regime" and "correlation"	Drop as a headline; keep as robustness
C	Use equity-heavy universes where a correlation channel exists	Channel exists, but D vs B is mixed in sign	Optional secondary universe

Proposed confirmatory protocol (for your approval)

- Fix **one primary specification** before looking at 2020-2025: SPY/IEF/GLD equal weight, the 2-state SPY-return HMM, D vs B at the 5% level. Candidate secondary specifications (listed and logged in advance): the global-equity universe, and the correlation-defined regime.
- Include **GARCH-t and DCC-t** as comparators, and report **power** next to every result.
- Treat 2016-2019 as the pseudo-holdout during development, and look at 2020-2025 once.

Questions for you

1. Is it acceptable to centre the thesis on **detectability, with a plain null if that is what the data say**? (If not, which positive-result path do you prefer?)
2. Do you approve the **single pre-registered confirmatory specification** above, or would you pick a different one?

3. Should the fund-based **1986 sample** appear only as supplementary evidence (proxy-quality caveats: mutual funds, once-a-day prices, long Treasuries rather than IEF)?
4. Should the universe stay **multi-asset**, or should an equity-heavy universe be added as a secondary test, knowing the Forbes-Rigobon caveat?

7. Revised plan if you agree (4 weeks)

Week	Deliverable
1	Your decisions above; freeze and log the specification; extend data to 2025 for the chosen universe only
2	Run the confirmatory test once on 2020-2025; include the 2022 episode analysis
3	Power and robustness analysis (window and refit sensitivity, benchmark comparison)
4	Write-up: results chapter, literature positioning, full reproducibility pass

8. Caveats to keep in mind

- Exploration results are in-sample and descriptive (smoothed regime weights); block-bootstrap intervals ignore HMM estimation uncertainty.
- About 214 correlated tests were run; only the Holm-corrected headline family should be quoted as evidence.
- Forecast samples overlap heavily across universes, so the tests are not independent.
- The HMM is the main regime model; the backtested alternatives are simple rules and variants, not an exhaustive search.
- The VaR/ES engine uses a Gaussian mixture against fat-tailed data. Power numbers are for that setup.

Sources: `output/PILOT_REPORT.md`, `output/explore/EXPLORATION_SUMMARY.md`, `output/explore/backtest_*.csv`. *Study guide for the underlying concepts:* `output/STUDY_GUIDE.pdf`.